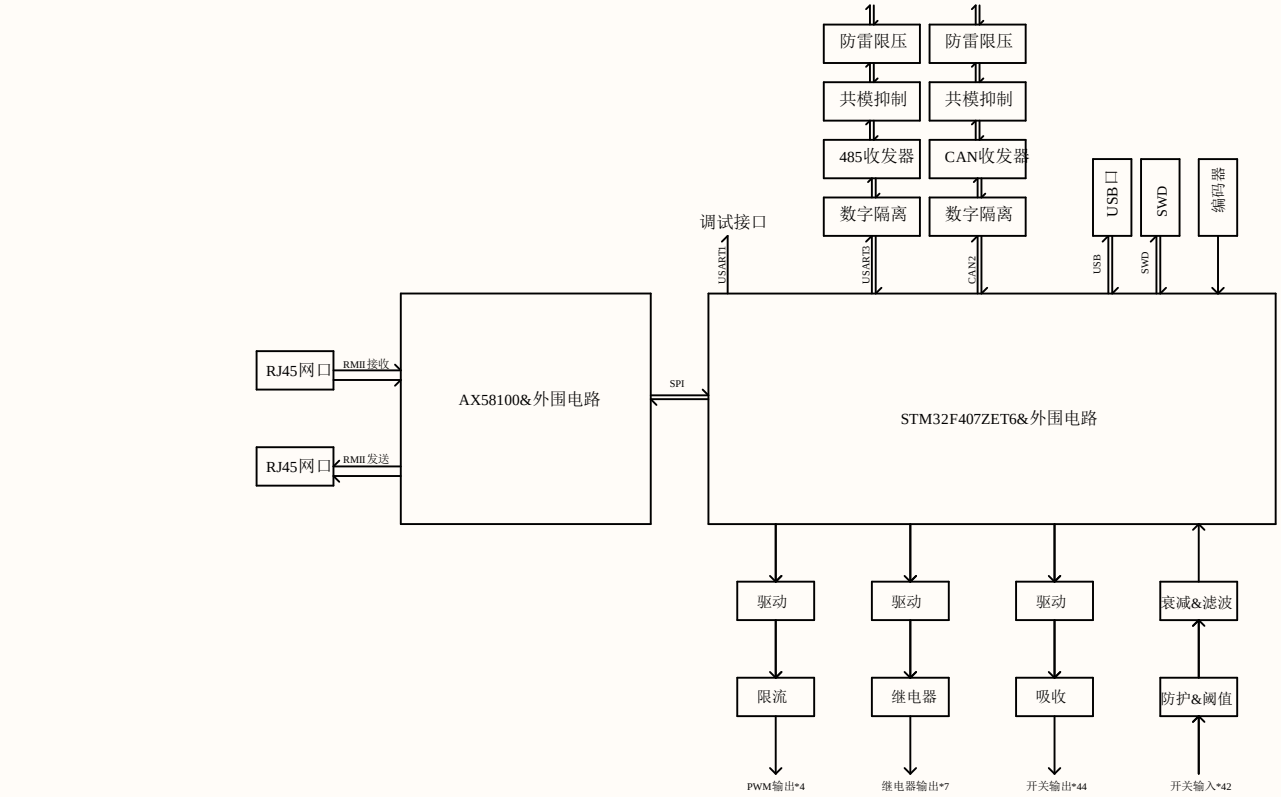
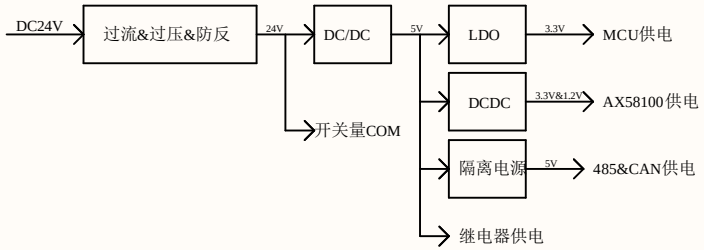
版本号	日期	修改人	修改内容
V1.0	2025.05.14	彭福权	新建

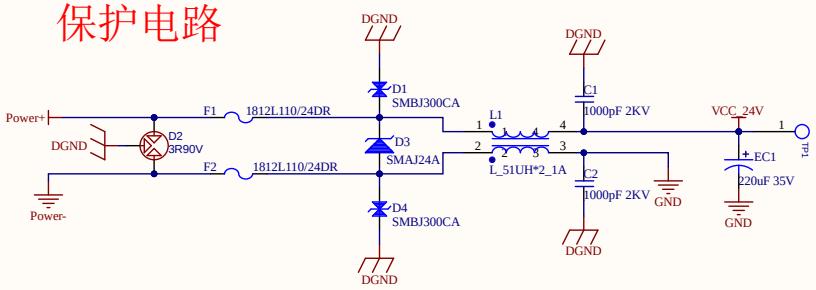
Title		
Size A4	Number	Revision
Date: 6/05/2025		Sheet of
File: E:\项目\..\00_Modify Note.SchDoc		Drawn By:

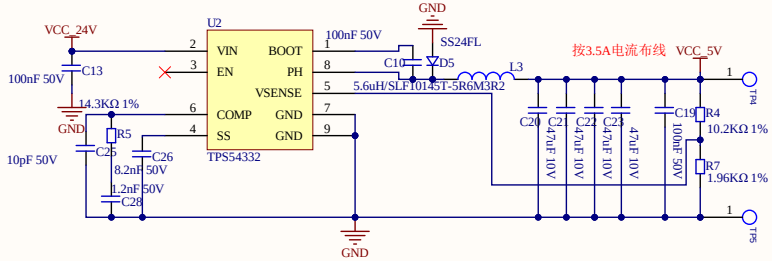




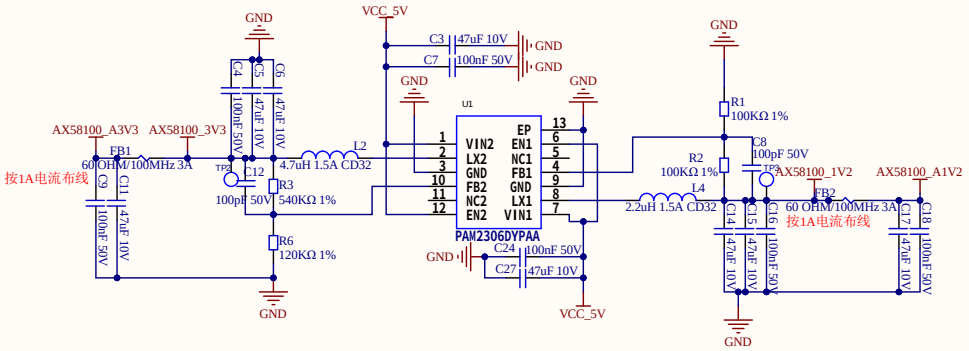
Title		
Size A4	Number	Revision
Date: 6/05/2025		Sheet of
File: E:\项目\..\01_BLOCK DIAGRAM.SchDoc		Drawn By:

保护电路

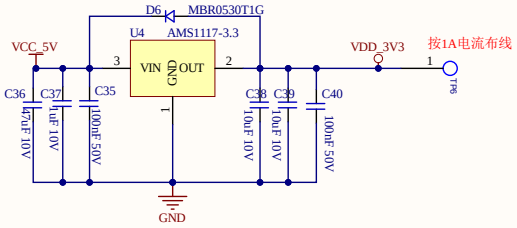




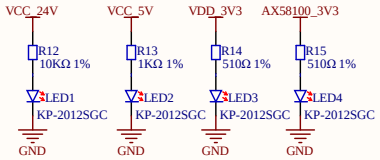
AX58100供电



单片机供电

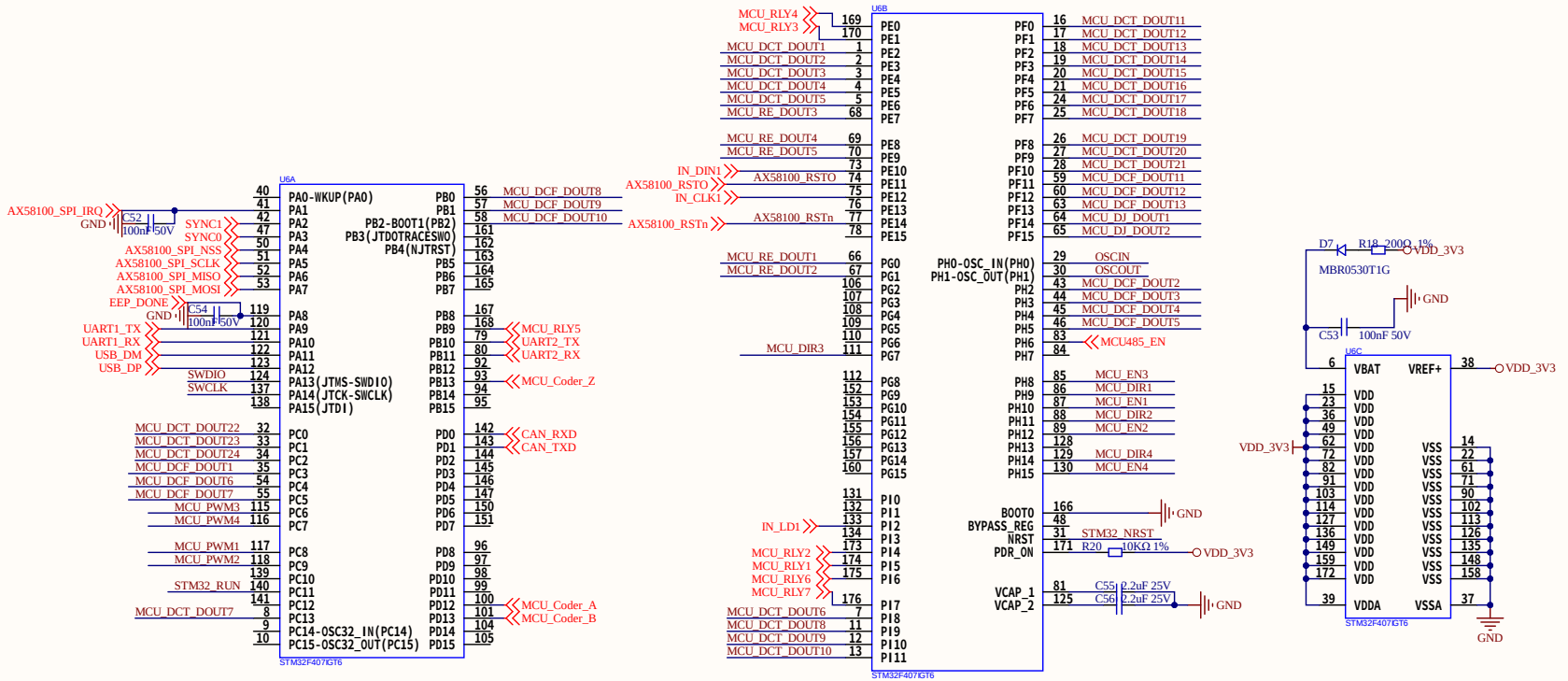


信号指示灯

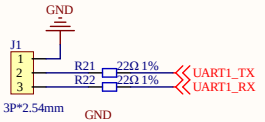


MARK1 MARK2 MARK3

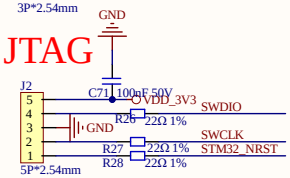
Title		
Size: A3	Number	Revision
Date: 6/05/2025	Sheet of	
File: E:\项目\项目1\包装通道1\包装通道1\02_POWER.SchDoc	Drawn By:	



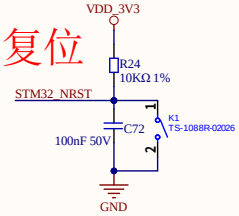
调试串口



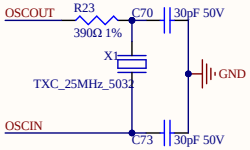
JTAG



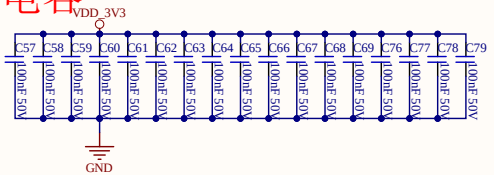
复位

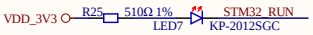


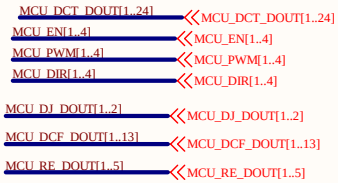
时钟

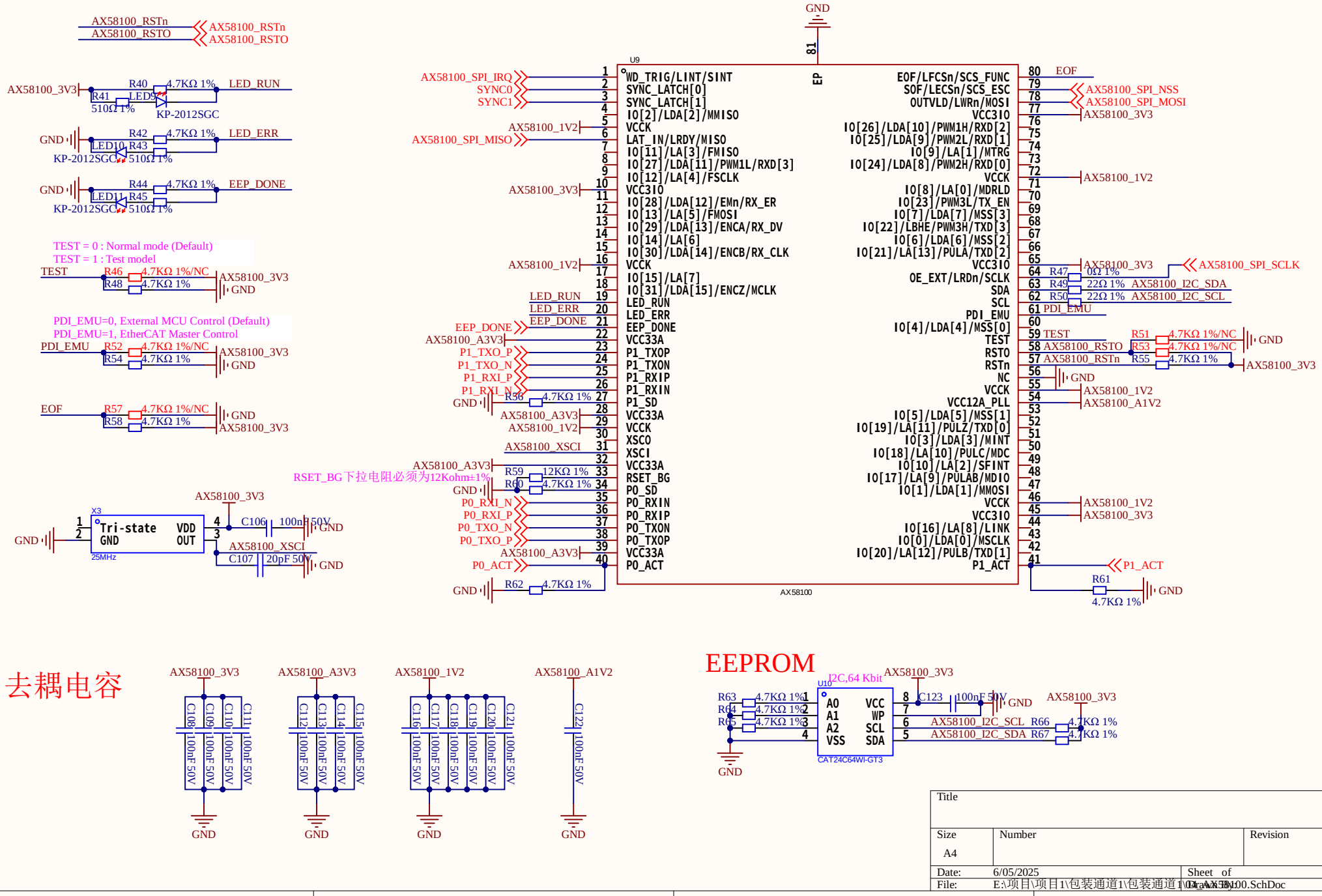


去耦电容

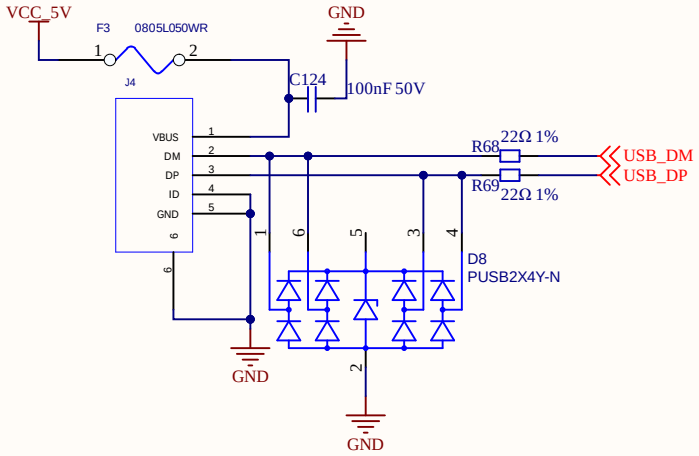




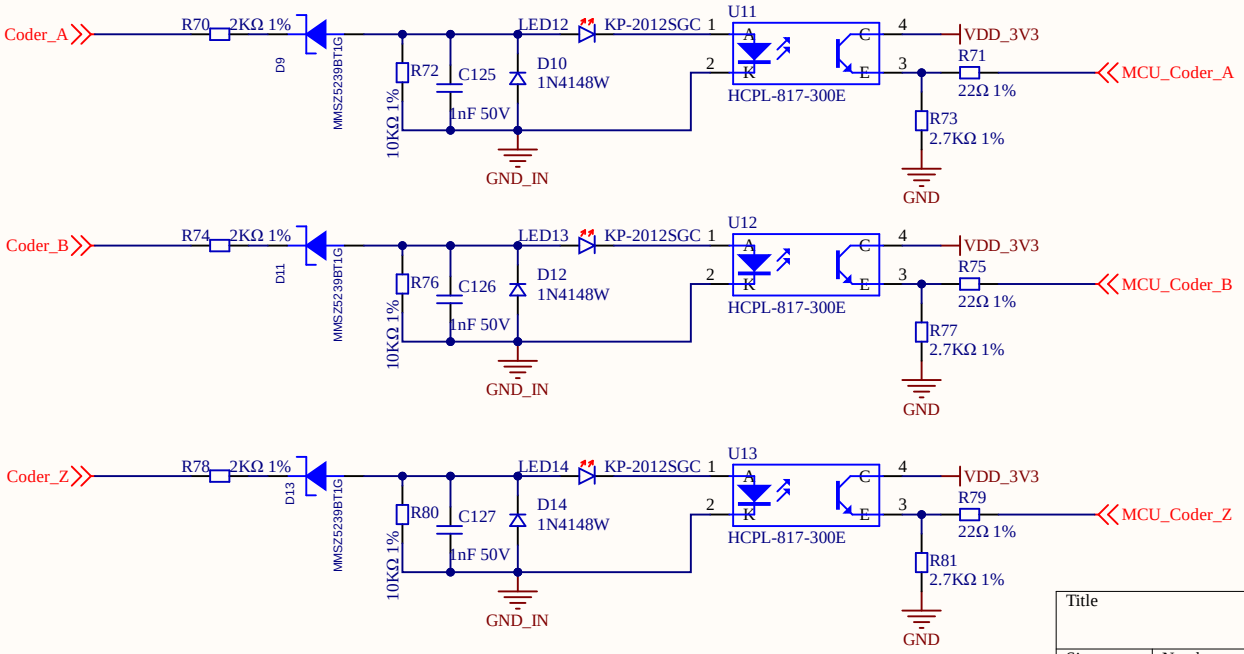



AX58100_RSTn
AX58100_RSTO
AX58100_3V3
R40 4.7KΩ 1%
LED_RUN
R41 510Ω 1%
LED9
KP-2012SGC
GND
R42 4.7KΩ 1%
LED_ERR
LED10
R43 510Ω 1%
R44 4.7KΩ 1%
EEP_DONE
LED11
R45 510Ω 1%
TEST = 0 : Normal mode (Default)
TEST = 1 : Test model
TEST
R46 4.7KΩ 1%/NC
R48 4.7KΩ 1%
PDI_EMU=0, External MCU Control (Default)
PDI_EMU=1, EtherCAT Master Control
PDI_EMU
R52 4.7KΩ 1%/NC
R54 4.7KΩ 1%
EOF
R57 4.7KΩ 1%/NC
R58 4.7KΩ 1%
X3
Tri-state
GND
VDD
OUT
25MHz
C106 100nF 50V
C107 20pF 50V
AX58100_XSCI
RSET_BG 下拉电阻必须为12Kohm±1%
U9
AX58100
WD_TRIG/LINT/SINT
SYNC_LATCH[0]
SYNC_LATCH[1]
IO[2]/LDA[2]/MMISO
VCCK
LAT_IN/LRDY/MISO
IO[11]/LA[3]/FMISO
IO[27]/LDA[11]/PWM1L/RXD[3]
IO[12]/LA[4]/FSCLK
VCC3IO
IO[28]/LDA[12]/EMn/RX_ER
IO[13]/LA[5]/FMOSI
IO[29]/LDA[13]/ENCA/RX_DV
IO[14]/LA[6]
IO[30]/LDA[14]/ENCB/RX_CLK
IO[15]/LA[7]
IO[31]/LDA[15]/ENCZ/MCLK
LED_RUN
LED_ERR
EEP_DONE
VCC33A
P1_TXOP
P1_TXON
P1_RXIP
P1_RXIN
P1_SD
RSET_BG
P0_SD
P0_RXIN
P0_RXIP
P0_TXON
P0_TXOP
P0_ACT
EP
EOF/LFCSn/SCS_FUNC
SOF/LECSn/SCS_ESC
OUTVLD/LWRn/MOSI
IO[26]/LDA[10]/PWM1H/RXD[2]
IO[25]/LDA[9]/PWM2L/RXD[1]
IO[9]/LA[1]/MTRG
IO[24]/LDA[8]/PWM2H/RXD[0]
IO[8]/LA[0]/MDRLD
IO[23]/PWM3L/TX_EN
IO[7]/LDA[7]/MSS[3]
IO[22]/LBHE/PWM3H/TXD[3]
IO[6]/LDA[6]/MSS[2]
IO[21]/LA[13]/PULA/TXD[2]
IO[4]/LDA[4]/MSS[0]
TEST
RSTO
RSTn
NC
VCCK
VCC12A_PLL
IO[5]/LDA[5]/MSS[1]
IO[19]/LA[11]/PULZ/TXD[0]
IO[3]/LDA[3]/MINT
IO[18]/LA[10]/PULC/MDC
IO[10]/LA[2]/SFINT
IO[17]/LA[9]/PULAB/MDIO
IO[1]/LDA[1]/MMOSI
VCC3IO
IO[16]/LA[8]/LINK
IO[0]/LDA[0]/MSCLK
IO[20]/LA[12]/PULB/TXD[1]
P1_ACT
R59 12KΩ 1%
R60 4.7KΩ 1%
R36 4.7KΩ 1%
R62 4.7KΩ 1%
R47 0Ω 1%
R49 22Ω 1%
R50 22Ω 1%
AX58100_SPI_SCLK
AX58100_I2C_SDA
AX58100_I2C_SCL
R51 4.7KΩ 1%/NC
R53 4.7KΩ 1%/NC
R55 4.7KΩ 1%
R61 4.7KΩ 1%
去耦电容
C108 100nF 50V
C109 100nF 50V
C110 100nF 50V
C111 100nF 50V
C112 100nF 50V
C113 100nF 50V
C114 100nF 50V
C115 100nF 50V
C116 100nF 50V
C117 100nF 50V
C118 100nF 50V
C119 100nF 50V
C120 100nF 50V
C121 100nF 50V
C122 100nF 50V
EEPROM
U10
I2C,64 Kbit
CAT24C64WI-GT3
A0
A1
A2
VSS
VCC
WP
SCL
SDA
R63 4.7KΩ 1%
R64 4.7KΩ 1%
R65 4.7KΩ 1%
C123 100nF 50V
R66 4.7KΩ 1%
R67 4.7KΩ 1%
Title
Size A4
Number
Revision
Date: 6/05/2025
Sheet of
File: E:\项目\项目1\包装通道1\包装通道1\Draw1.SchDoc

USB

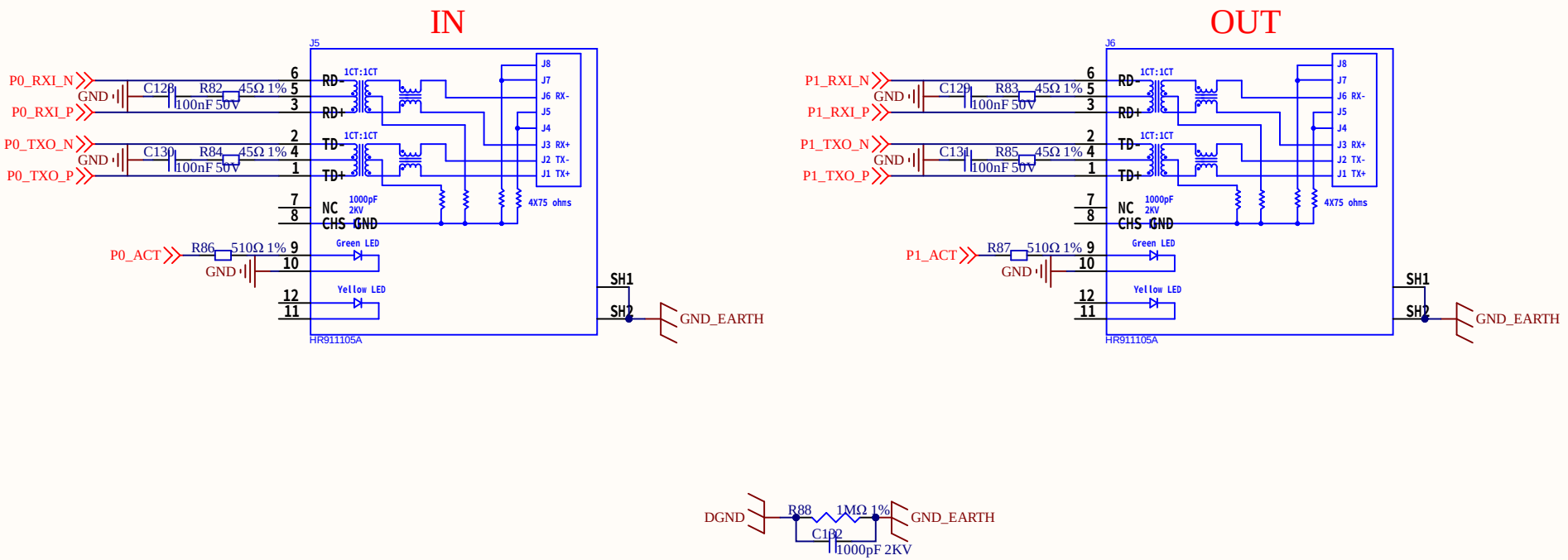


Coder

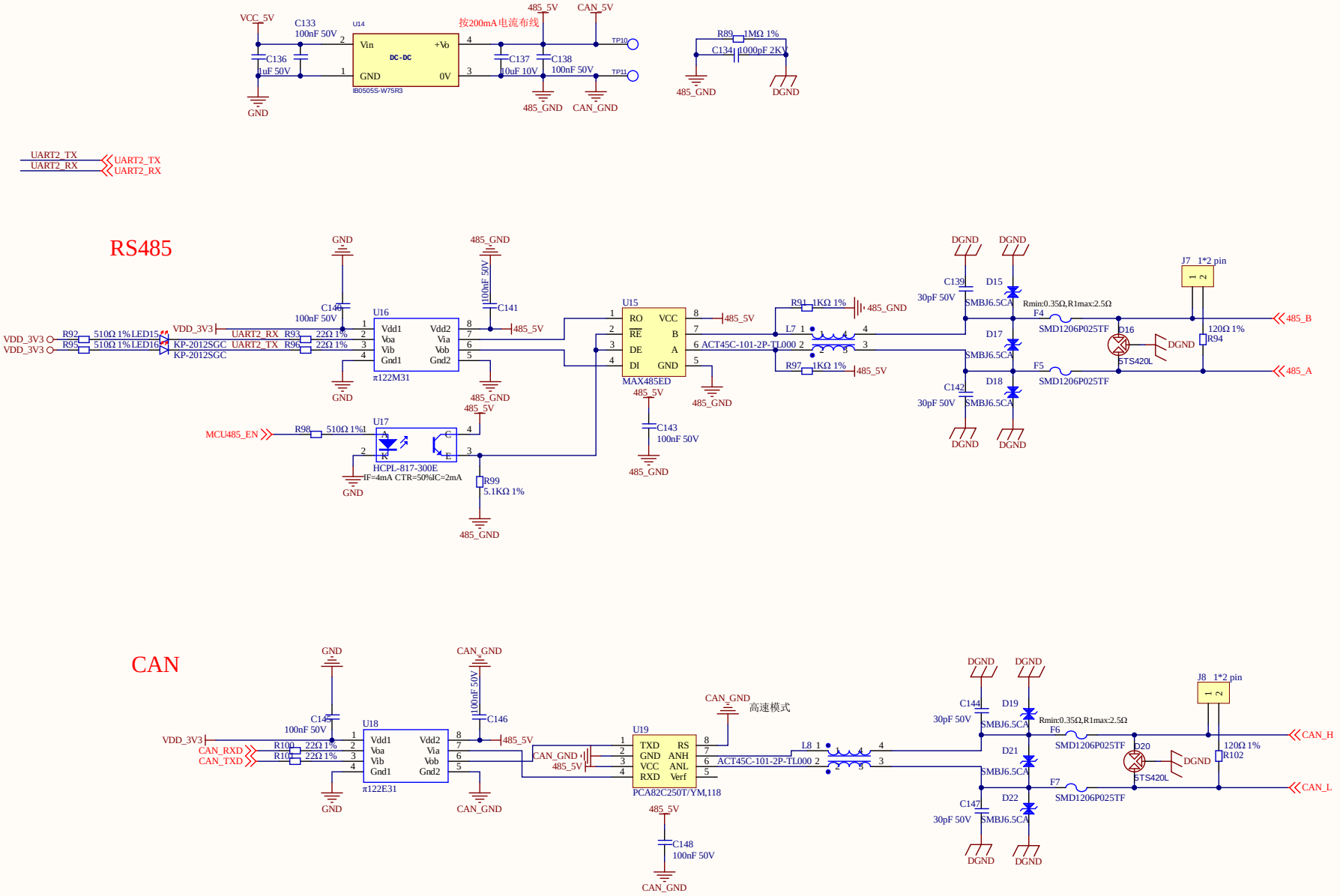


Title		
Size: A4	Number	Revision
Date: 6/05/2025		Sheet of
File: E:\项目\..\05_USB&Coder.SchDoc		Drawn By:

网口



Title		
Size A4	Number	Revision
Date: 6/05/2025		Sheet of
File: E:\项目\项目1\包装通道1\包装通道1\Drawings\		Drawn By:

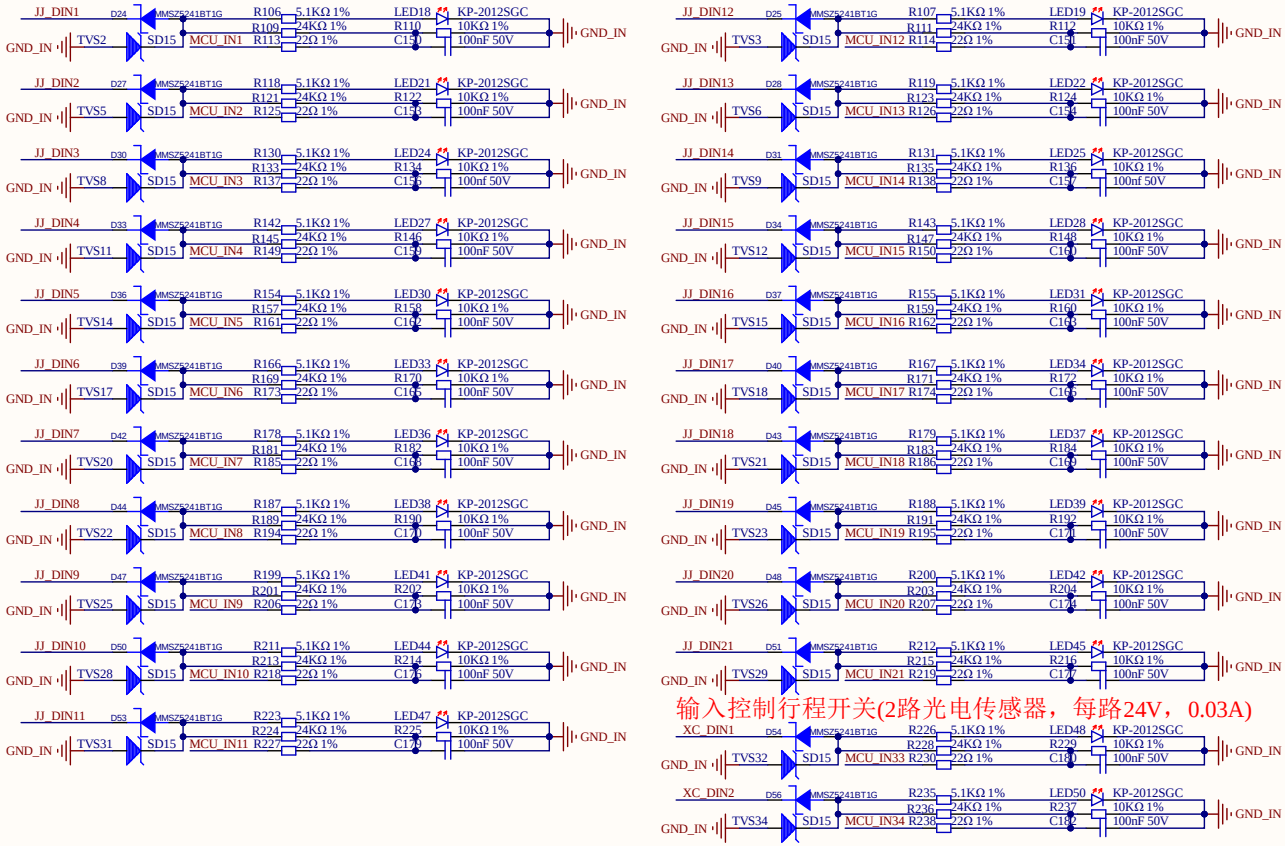


Title		
Size A3	Number	Revision
Date: 6/05/2025		Sheet of
File: E:\项目\..\07_RS485&CAN.SchDoc		Drawn By:

共42路输入

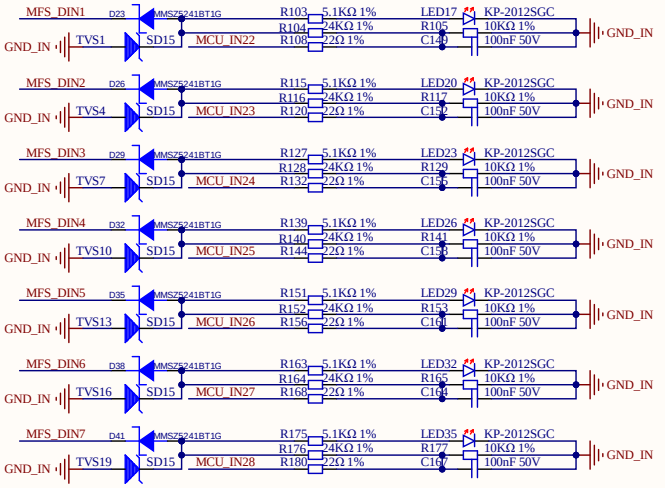
检测接近开关输入(21路XS112BHPAL2，24V，0.1A)

稳压管选择11V，LED工作电流约为2.2mA,输入为24V时，输出为(24-0.7-11)*10/(10+24)=3.6V。

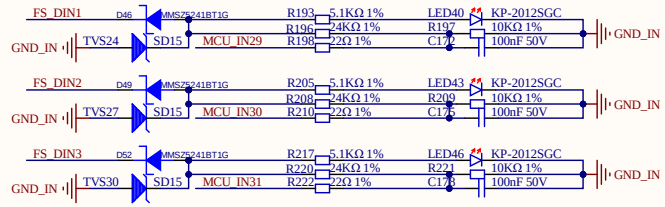


输入控制行程开关(2路光电传感器，每路24V，0.03A)

检测传感器输入(漫反射，7路XUM8APSBL2，24V)



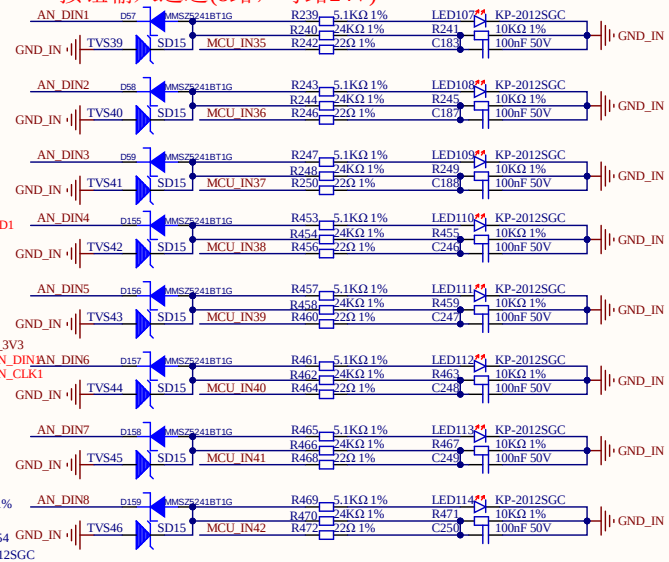
检测传感器输入(反射，3路XUM9APXBL2，24V)

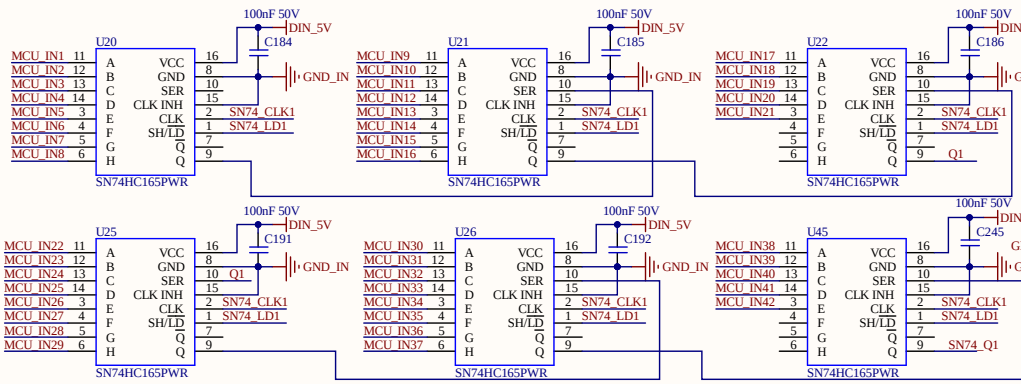


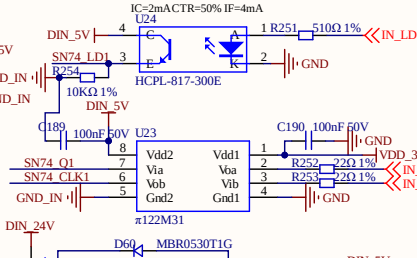
门板安全开关输入(1路XCSDMC5902，24V)

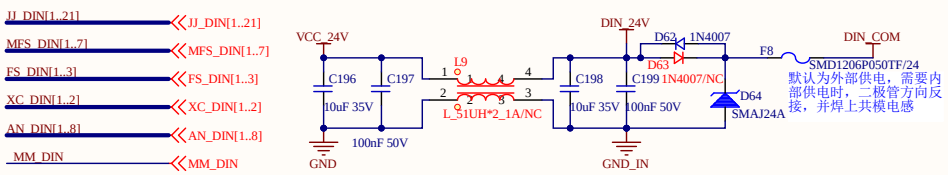


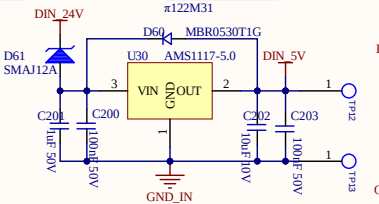
按钮输入通道(8路，每路24V)

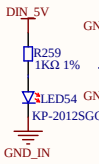






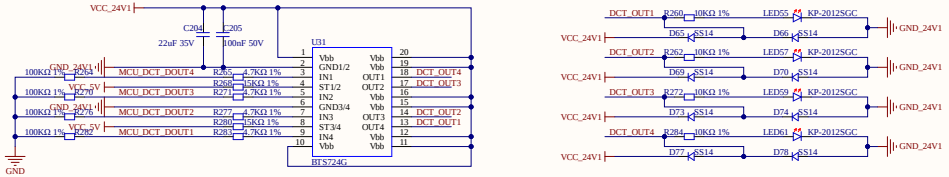


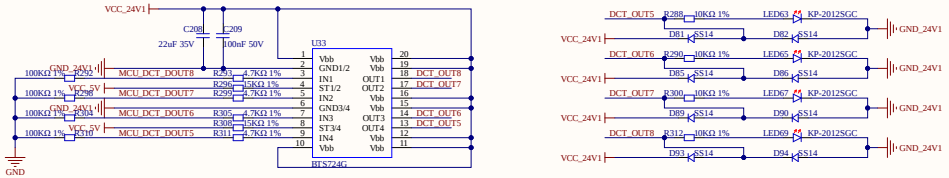


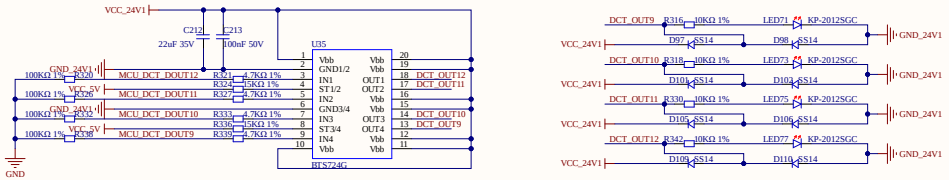


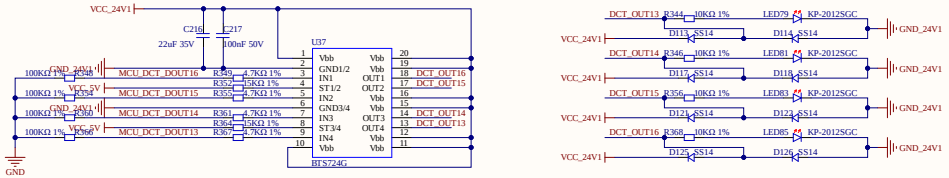
共44路输出

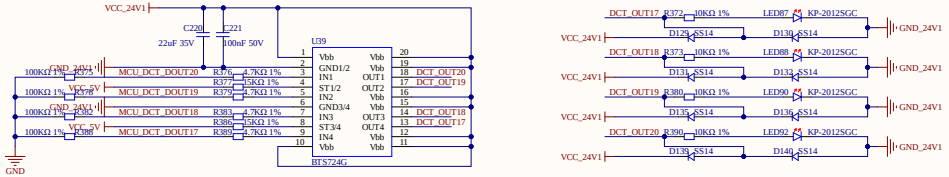
输出控制电磁铁(24路，每路24V，0.49A)

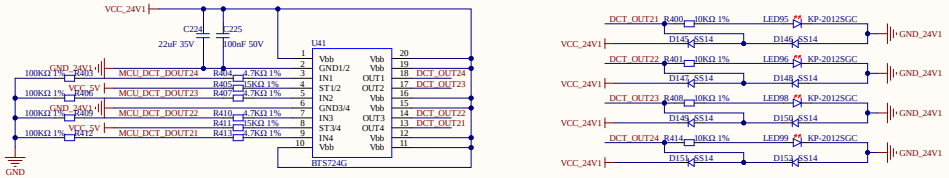








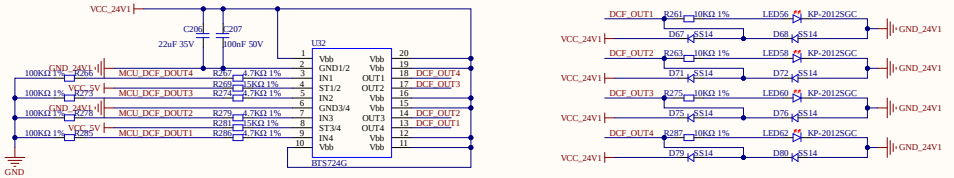


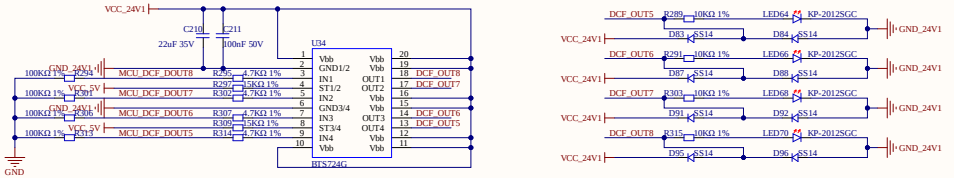


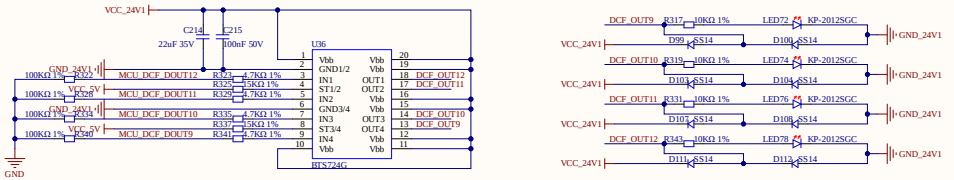




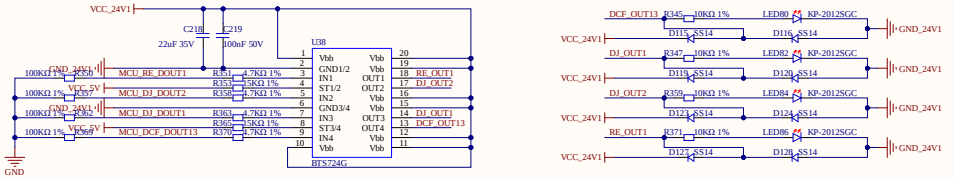
输出控制打开气缸(13路电磁阀，每路24V，0.125A)



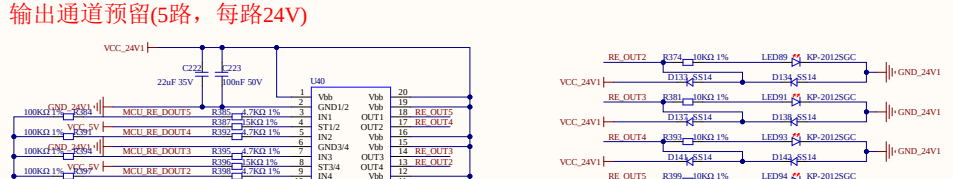


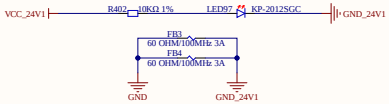


输出控制电机(2路，每路24V)



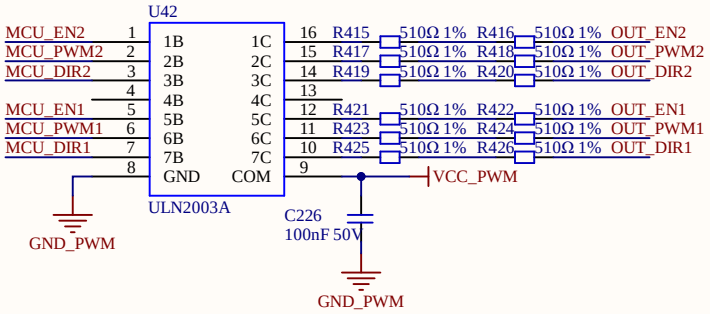
输出通道预留(5路，每路24V)

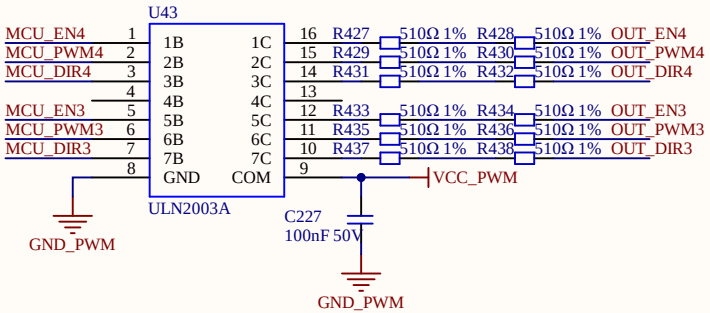


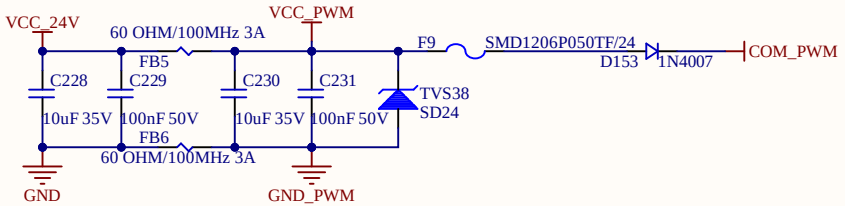


Title		
Size: A2	Number	Revision
Date: 6/05/2025	Sheet of	
File: E:\项目\项目1\包装通道1\包装通道1.SchDoc	Drawn By:	

4路PWM输出





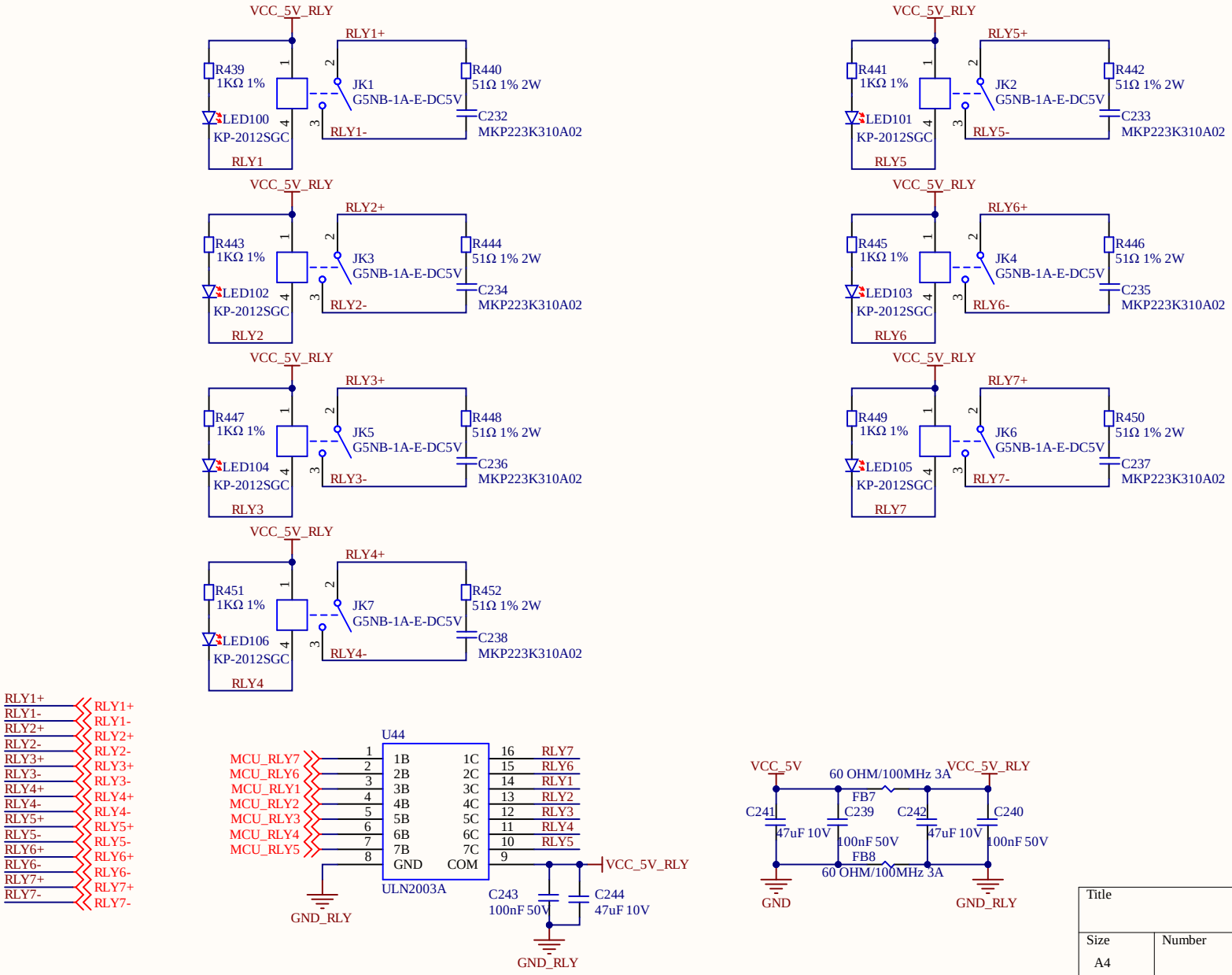


MCU_EN[1..4] <<MCU_EN[1..4]
MCU_PWM[1..4] <<MCU_PWM[1..4]
MCU_DIR[1..4] <<MCU_DIR[1..4]

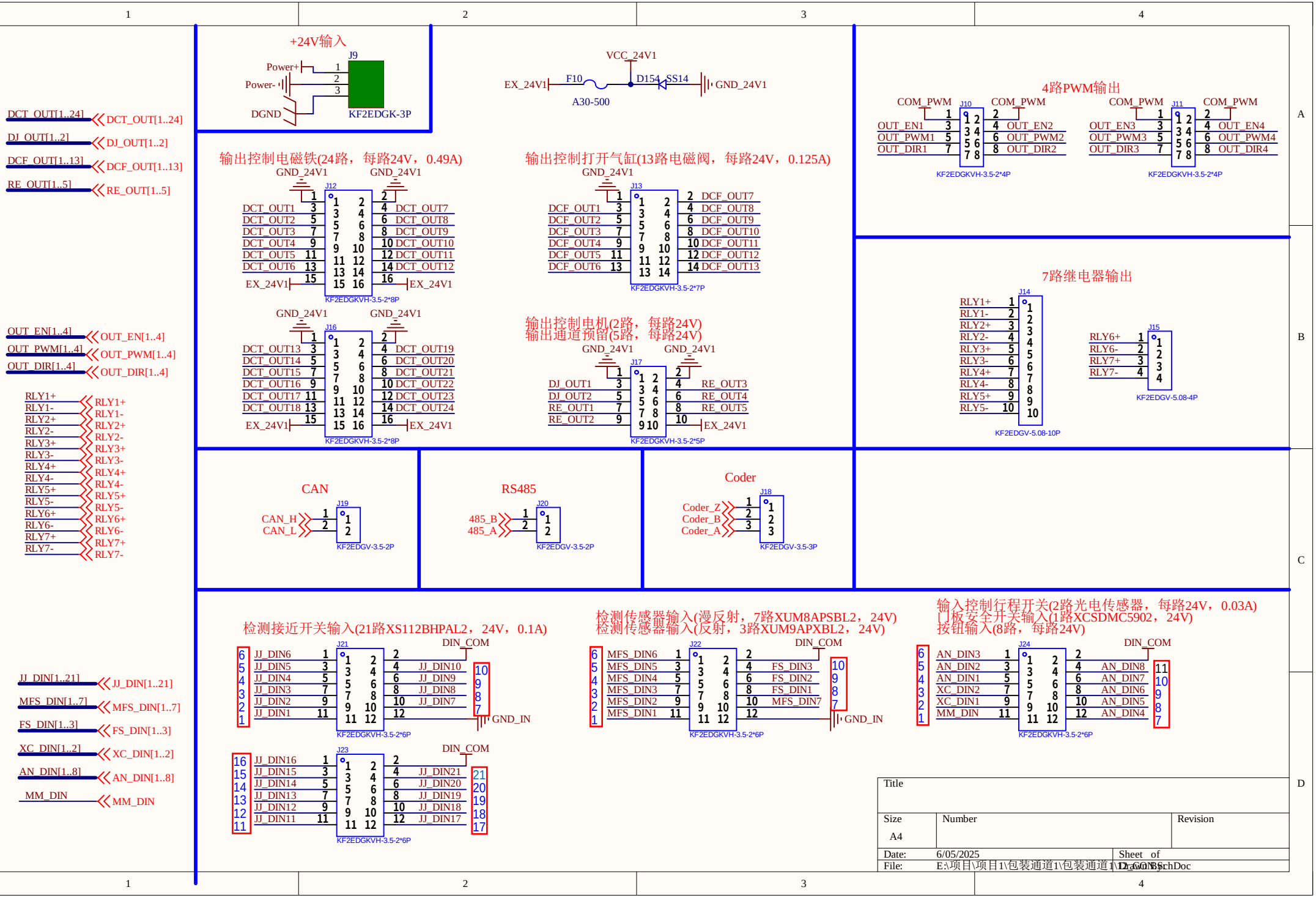
OUT_EN[1..4] <<OUT_EN[1..4]
OUT_PWM[1..4] <<OUT_PWM[1..4]
OUT_DIR[1..4] <<OUT_DIR[1..4]

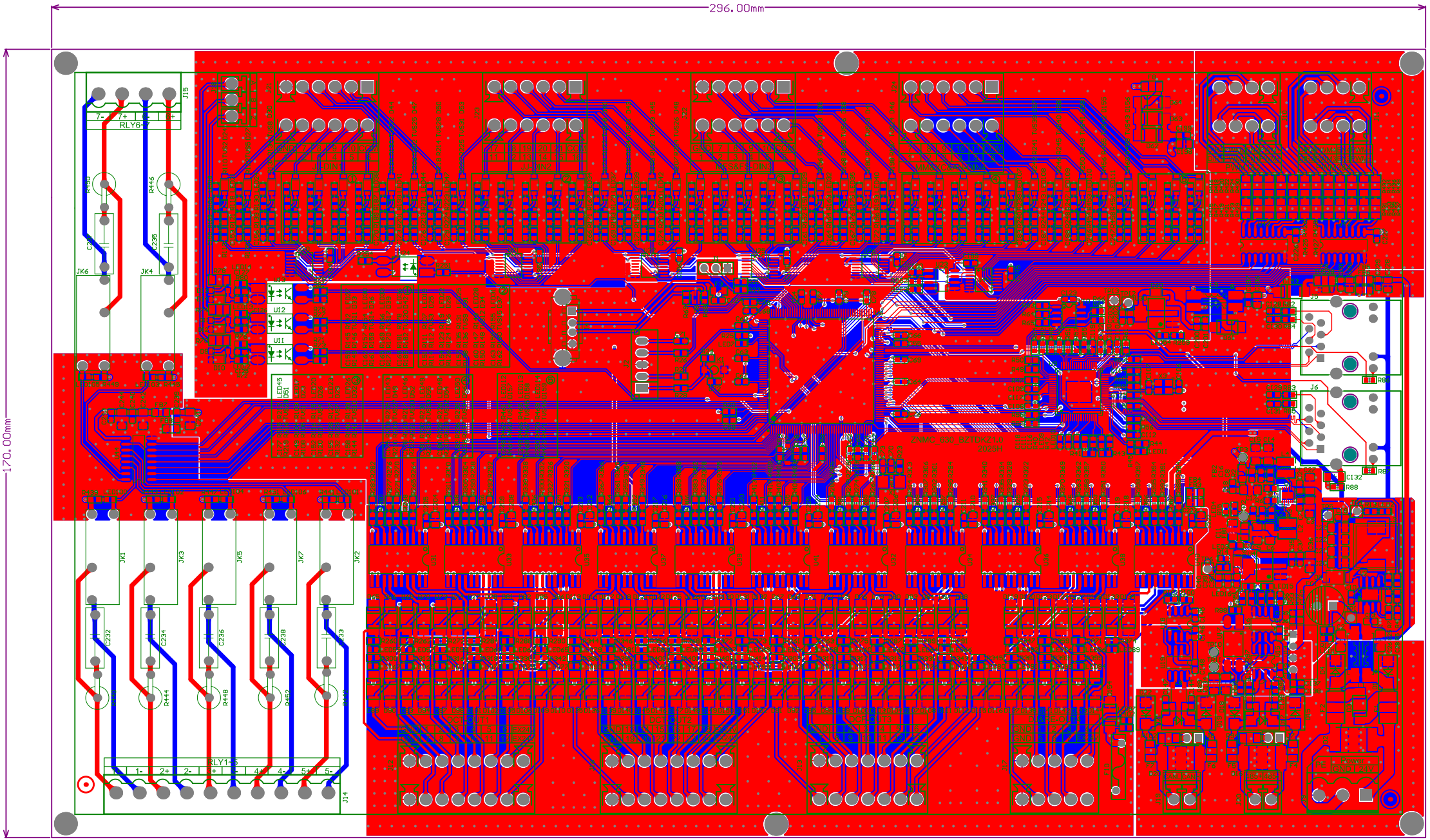
Title		
Size A4	Number	Revision
Date: 6/05/2025		Sheet of
File: E:\项目\项目1\包装通道1\包装通道1\DrawMyOUT.SchDoc		Drawn By:

7路继电器输出



Title		
Size A4	Number	Revision
Date: 6/05/2025		Sheet of
File: E:\项目\项目1\包装通道1\包装通道1\Drawing.SchDoc		Drawn By:


+24V输入
J9
Power+
Power-
DGND
KF2EDGK-3P
VCC_24V1
EX_24V1
F10
A30-500
D154
SS14
GND_24V1
4路PWM输出
COM_PWM
J10
J11
OUT_EN1
OUT_PWM1
OUT_DIR1
OUT_EN2
OUT_PWM2
OUT_DIR2
OUT_EN3
OUT_PWM3
OUT_DIR3
OUT_EN4
OUT_PWM4
OUT_DIR4
KF2EDGKVH-3.5-2*4P
输出控制电磁铁(24路，每路24V，0.49A)
J12
DCT_OUT1
DCT_OUT2
DCT_OUT3
DCT_OUT4
DCT_OUT5
DCT_OUT6
DCT_OUT7
DCT_OUT8
DCT_OUT9
DCT_OUT10
DCT_OUT11
DCT_OUT12
EX_24V1
KF2EDGKVH-3.5-2*8P
输出控制打开气缸(13路电磁阀，每路24V，0.125A)
J13
DCF_OUT1
DCF_OUT2
DCF_OUT3
DCF_OUT4
DCF_OUT5
DCF_OUT6
DCF_OUT7
DCF_OUT8
DCF_OUT9
DCF_OUT10
DCF_OUT11
DCF_OUT12
DCF_OUT13
KF2EDGKVH-3.5-2*7P
7路继电器输出
J14
RLY1+
RLY1-
RLY2+
RLY2-
RLY3+
RLY3-
RLY4+
RLY4-
RLY5+
RLY5-
J15
RLY6+
RLY6-
RLY7+
RLY7-
KF2EDGV-5.08-10P
KF2EDGV-5.08-4P
输出控制电机(2路，每路24V)
输出通道预留(5路，每路24V)
J16
DCT_OUT13
DCT_OUT14
DCT_OUT15
DCT_OUT16
DCT_OUT17
DCT_OUT18
DCT_OUT19
DCT_OUT20
DCT_OUT21
DCT_OUT22
DCT_OUT23
DCT_OUT24
J17
DJ_OUT1
DJ_OUT2
RE_OUT1
RE_OUT2
RE_OUT3
RE_OUT4
RE_OUT5
CAN
J19
CAN_H
CAN_L
KF2EDGV-3.5-2P
RS485
J20
485_B
485_A
Coder
J18
Coder_Z
Coder_B
Coder_A
KF2EDGV-3.5-3P
检测接近开关输入(21路XS112BHPAL2，24V，0.1A)
J21
J23
JJ_DIN1
JJ_DIN2
JJ_DIN3
JJ_DIN4
JJ_DIN5
JJ_DIN6
JJ_DIN7
JJ_DIN8
JJ_DIN9
JJ_DIN10
JJ_DIN11
JJ_DIN12
JJ_DIN13
JJ_DIN14
JJ_DIN15
JJ_DIN16
JJ_DIN17
JJ_DIN18
JJ_DIN19
JJ_DIN20
JJ_DIN21
DIN_COM
GND_IN
KF2EDGKVH-3.5-2*6P
检测传感器输入(漫反射，7路XUM8APSBL2，24V)
检测传感器输入(反射，3路XUM9APXBL2，24V)
J22
MFS_DIN1
MFS_DIN2
MFS_DIN3
MFS_DIN4
MFS_DIN5
MFS_DIN6
MFS_DIN7
FS_DIN1
FS_DIN2
FS_DIN3
输入控制行程开关(2路光电传感器，每路24V，0.03A)
门板安全开关输入(1路XCSDMC5902，24V)
按钮输入(8路，每路24V)
J24
MM_DIN
XC_DIN1
XC_DIN2
AN_DIN1
AN_DIN2
AN_DIN3
AN_DIN4
AN_DIN5
AN_DIN6
AN_DIN7
AN_DIN8
DCT_OUT[1..24]
DJ_OUT[1..2]
DCF_OUT[1..13]
RE_OUT[1..5]
OUT_EN[1..4]
OUT_PWM[1..4]
OUT_DIR[1..4]
RLY1+
RLY1-
RLY2+
RLY2-
RLY3+
RLY3-
RLY4+
RLY4-
RLY5+
RLY5-
RLY6+
RLY6-
RLY7+
RLY7-
JJ_DIN[1..21]
MFS_DIN[1..7]
FS_DIN[1..3]
XC_DIN[1..2]
AN_DIN[1..8]
MM_DIN
Title
Size
A4
Number
Revision
Date:
6/05/2025
Sheet of
File:
E:\项目\项目1\包装通道1\包装通道1\Drawing.SchDoc
Drawn By:


296.00mm
170.00mm
J15
RLY6-7
JK6
JK4
U12
U11
LED45
D51
JK1
JK3
JK5
JK7
JK2
RLY1-5
J14
J10
J6
C132
R88
ZYNQ_800_BZTDXZLG
2024